

Product Requirements Document

Backlink Exchange Platform

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1. Executive Summary

This platform is a two-sided marketplace for SEO backlink acquisition. Website owners can exchange backlinks with each other using a credit-based economy, or purchase backlinks outright through a paid marketplace. The platform handles discovery, outreach, request management, link verification, and credit settlement — replacing ad-hoc email outreach with a structured, trust-scored workflow.

2. Problem Statement

For website owners seeking backlinks

- Finding relevant, high-quality sites to exchange links with is time-consuming and manual
- Cold email outreach has low response rates with no accountability
- There is no standard way to track the status of outreach or verify that agreed links went live
- Buying links directly is risky — no verification, no recourse

For website owners receiving link requests

- Inbound requests arrive unstructured across email and contact forms
- No central place to review, accept, or decline requests
- No incentive to respond quickly or at all
- No way to monetise their link placements efficiently

Platform opportunity

A structured, trust-scored exchange removes friction on both sides — requesters get access to a qualified inventory, publishers get organised inbound requests and earn credits they can redeem for their own link placements.

3. Goals & Success Metrics

Business Goals

- Build a sustainable credit-based economy where exchange is the default and marketplace purchases are upsell
- Achieve platform liquidity: enough supply (publishers) and demand (requesters) that most requests get a response within 7 days
- Generate revenue through credit purchases and marketplace transactions

Success Metrics

Metric	Target (6 months)
Registered users	5,000
Verified projects	8,000
Requests sent per month	15,000
Request acceptance rate	≥ 35%
Median time to first response	≤ 5 days
Credits purchased (MRR)	\$20,000
Marketplace GMV	\$10,000/month
Average responsiveness score	≥ 65%
Month-1 retention	≥ 40%

4. User Personas

4.1 The SEO Specialist ("Alex")

- Works at a digital agency managing 10–30 client websites
- Needs to build links at scale efficiently
- Values data: DR, DA, TF, traffic before deciding to pursue a link
- Frustrated by: slow replies, broken links, no tracking
- Uses: Ahrefs, SEMrush, outreach tools
- Primary action: sends many outgoing requests, monitors their status, exports reports for clients

4.2 The Indie Founder ("Priya")

- Runs 1–3 niche content sites, doing SEO herself with limited time
- Wants quality over quantity
- Primary action: lists her sites in the exchange, selectively accepts incoming requests, occasionally sends outgoing ones
- Values: responsiveness score (wants to look trustworthy), simple UX

4.3 The Content Publisher ("Marcus")

- Runs a high-DA media site (DA 60+), receives many inbound link requests
- Wants to monetise placements — marketplace listing is appealing

- Primary action: reviews and accepts/rejects incoming requests, sets turnaround times, marks links live to earn credits or cash
- Values: clean request inbox, ability to set guidelines upfront

4.4 The Growth Marketer ("Sofia")

- Works in-house at a SaaS startup, needs backlinks fast for a product launch
 - Has budget, will purchase credits or marketplace links
 - Primary action: browses exchange, buys marketplace links, monitors outgoing requests
 - Values: speed, quality metrics, link verification
-

5. User Journeys

5.1 Requester Flow

```

Sign up → Verify email
→ Add project (enter domain + name)
→ Verify domain ownership (meta tag or DNS)
→ Browse Exchange tab (filter by category, metrics, language, country)
→ Find a suitable site → view metrics, guidelines, responsiveness score
→ Send request (choose type, fill details, confirm credit cost)
→ Credits deducted → Request sent (Pending)
→ Publisher accepts → Notified (Accepted, TAT shown)
→ Publisher marks Live → Notified (Live)

```

5.2 Publisher Flow

```

Sign up → Add project → Verify domain
→ Enable Exchange → Set guidelines
→ Receive incoming requests (email + in-app)
→ Review: requester metrics, source URL, anchor text, target URL
→ Accept (set TAT) / Reject / Put on Hold
→ Publish the link on site
→ Mark as Live → Link verified → Credits earned

```

5.3 Marketplace Buyer Flow

```

Browse Marketplace → filter by price, metrics, category
→ Select website → view listing details
→ Click Buy → provide link details
→ Pay via Stripe
→ Publisher fulfils order → Link goes live → Buyer notified

```

5.4 Credit Top-Up Flow

```

Running low on credits (≤ 2 remaining) → Warning shown
→ Click Credits badge in header → "Add Credits"
→ Choose package (Starter / Growth / Pro / Agency)
→ Enter payment details → Pay via Stripe
→ Credits added instantly

```

6. Feature Requirements

6.1 Authentication & Accounts

#	Requirement	Priority
A1	Email + password registration with email verification	P0
A2	Login with JWT access token + httpOnly refresh token	P0
A3	Forgot password / reset password via email	P0
A4	Google OAuth login	P1
A5	Account deletion with data cleanup	P1
A6	Profile page: name, username, bio, website, location, avatar	P1
A7	Subscription plan shown on profile (Free / Pro / Agency)	P1

6.2 Projects

#	Requirement	Priority
PR1	Add a project by entering domain + display name	P0
PR2	Verify domain ownership via HTML meta tag	P0
PR3	Verify domain ownership via DNS TXT record	P1
PR4	View project list with SEO metrics (DA, DR, TF, Traffic, Spam Score)	P0
PR5	Enable/disable Exchange participation per project	P0
PR6	Set Link Insertion and Guest Post guidelines per project	P0
PR7	Edit project display name	P0
PR8	Add notes to a project (private, internal)	P1
PR9	Remove a project	P0
PR10	SEO metrics auto-refreshed nightly via third-party APIs	P1
PR11	Metric delta view (change over 7 / 30 / 90 days)	P1
PR12	Per-project responsiveness score displayed	P0

6.3 Exchange — Website Browser

#	Requirement	Priority
EX1	Browse all exchange-enabled websites with metrics	P0
EX2	Search by domain name	P0

EX3	Filter by category, language, country, tags	P0
EX4	Filter by metric ranges (DA, DR, TF, Traffic, Spam Score)	P0
EX5	Filter by request type availability (Link Insertion / Guest Post)	P0
EX6	Sort by any metric column, ascending or descending	P0
EX7	Pagination (server-side)	P0
EX8	View website detail: guidelines, full metrics, responsiveness score	P0
EX9	Send request from website detail view	P0
EX10	Export website list (CSV, TSV, JSON, XLS)	P1
EX11	AI-powered suggestions: recommended sites based on user's project niche	P2

6.4 Exchange — Request Management

#	Requirement	Priority
RQ1	View all incoming requests: domain, metrics, type, status	P0
RQ2	View all outgoing requests with same columns	P0
RQ3	Request detail dialog: source URL, anchor text, target URL, title, description	P0
RQ4	Publisher actions: Accept (with TAT), Reject, Put on Hold	P0
RQ5	Publisher action: Mark as Live (triggers verification, earns credits)	P0
RQ6	Requester action: Mark as Live (for accepted outgoing requests)	P0
RQ7	Automated link verification on Mark as Live	P1
RQ8	Credits displayed per request (cost for outgoing, earn for incoming)	P0
RQ9	Status badge with colour coding	P0
RQ10	Filter by status, type, domain search	P0
RQ11	Sort by domain, DR, DA, TF, Traffic, RD, Spam Score	P0
RQ12	Cross-project view (all requests across all user's projects)	P0
RQ13	Per-project filtered view	P0
RQ14	Export requests (CSV, TSV, JSON, XLS)	P1
RQ15	Auto-expire Pending requests after 30 days, penalise responsiveness	P1

6.5 Request Creation

#	Requirement	Priority
SC1	Choose request type: Link Insertion or Guest Post	P0

SC2	Link Insertion fields: source URL, anchor text, target URL, description	P0
SC3	Guest Post fields: source URL (target blog), post title, description	P0
SC4	Show credit cost before confirming	P0
SC5	Block submission if insufficient credits (show top-up prompt)	P0
SC6	Block if active request already exists for same requester+publisher domain	P0
SC7	Notify publisher by email + in-app on submission	P0

6.6 Credits

#	Requirement	Priority
CR1	Credit balance always visible in top navigation	P0
CR2	Credits dropdown: balance, View Transactions, Add Credits	P0
CR3	Deduct credits when outgoing request is sent	P0
CR4	Refund credits if request is rejected	P0
CR5	Award credits to publisher when link is marked Live + verified	P0
CR6	Welcome bonus credits on registration	P0
CR7	Purchase credits via Stripe (4 packages)	P0
CR8	Full transaction history (date, description, amount, running balance)	P0
CR9	Low-credit warning when balance ≤ 2	P1
CR10	Credits never expire	P0

6.7 Responsiveness Score

#	Requirement	Priority
RS1	Account-level responsiveness score visible in top bar	P0
RS2	Score = % of incoming requests responded to (not Pending)	P0
RS3	Per-project responsiveness score	P0
RS4	Per-domain score visible to other users in the exchange browser	P0
RS5	Colour-coded: green $\geq 75\%$, amber $\geq 50\%$, red $< 50\%$	P0
RS6	Tooltip explaining score and how it affects platform visibility	P0
RS7	Auto-expired requests count as non-responded, lowering score	P1
RS8	Exchange sort boosts high-responsiveness sites	P1

6.8 Marketplace

#	Requirement	Priority
MK1	Browse paid listings with price, metrics, category	P0
MK2	Filter and sort same as exchange	P0
MK3	Purchase a placement via Stripe	P0
MK4	Order tracking: Pending → In Progress → Live	P0
MK5	Publisher fulfils order, marks live	P0
MK6	Order history for buyers	P0
MK7	Link verification on marketplace orders	P1
MK8	Refund if link not live within agreed window	P1

6.9 Dashboard

#	Requirement	Priority
DB1	Project selector (switch between projects)	P0
DB2	Request summary cards: Pending, Accepted, Rejected, On Hold counts	P0
DB3	Recent requests list (mini view with click-to-expand)	P0
DB4	Metric delta panel (DA, DR, TF, Traffic, Spam, Responsiveness)	P1
DB5	Date range picker for deltas (7 / 30 / 90 days)	P1
DB6	Per-project todo list	P1

6.10 Notifications

#	Requirement	Priority
NT1	Email on new incoming request	P0
NT2	Email on request accepted (with TAT)	P0
NT3	Email on request rejected	P0
NT4	Email on link verified live	P0
NT5	Email on credit purchase (receipt)	P0
NT6	Email warning when credits $\leq$ 2	P1
NT7	In-app notification bell (real-time via WebSocket / SSE)	P1
NT8	Weekly email digest of pending actions	P2

6.11 Settings

#	Requirement	Priority
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ST1	Change email	P1
ST2	Change password	P1
ST3	Email notification preferences per event type	P1
ST4	Connected OAuth accounts	P1
ST5	Delete account	P1

6.12 Plans & Upgrade

#	Requirement	Priority
PL1	Free plan with usage limits (3 projects, 10 requests/month)	P1
PL2	Pro plan: more projects, higher limits, advanced filters	P1
PL3	Agency plan: unlimited projects, team members, white-label exports	P2
PL4	Upgrade page with plan comparison	P1
PL5	Subscription managed via Stripe Billing	P1

7. Functional Specifications

7.1 Credit Calculation

Credits per link are calculated from publisher SEO metrics and normalised to a 1–10 scale:

```
dr_part      = floor(DR × 0.20)
da_part      = floor(DA × 0.15)
tf_part      = floor(TF × 0.10)
```

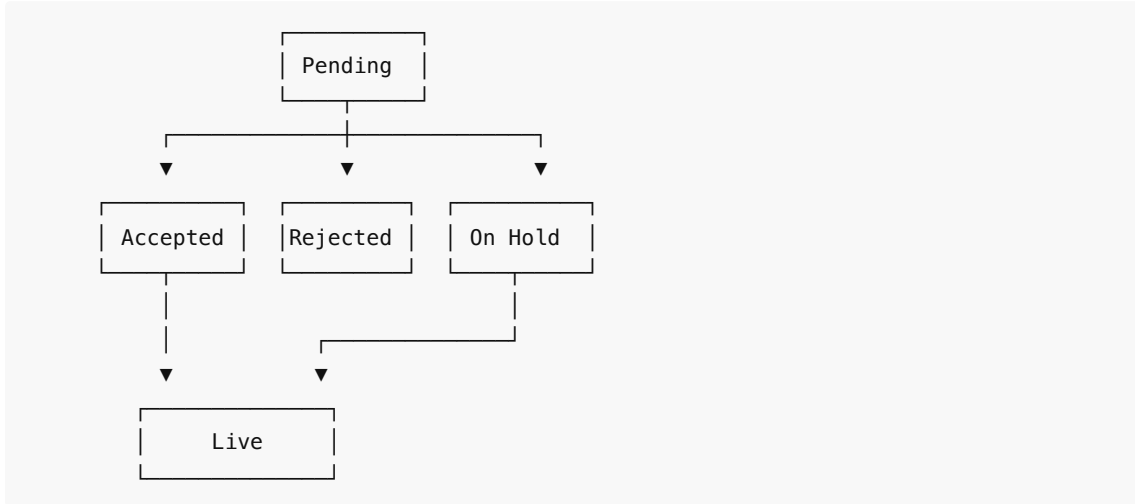
```
traffic_part:
  ≥ 10,000,000 → 18
  ≥  2,000,000 → 12
  ≥   500,000 →  8
  ≥   150,000 →  5
  ≥    50,000 →  3
  ≥     7,000 →  2
  default     →  1
```

```
spam_penalty:
  spam ≥ 15 → -6
  spam ≥  8 → -3
  spam ≥  4 → -1
  default  →  0
```

```
raw      = max(1, dr_part + da_part + tf_part + traffic_part - spam_penalty)
credits = max(1, min(10, round(1 + (raw - 1) / 59 × 9)))
```


Result is always **1–10 credits**. The same value is both the cost charged to the requester and the reward earned by the publisher.

7.2 Request Status Transitions



- **Rejected** → credits refunded to requester immediately
- **On Hold** → publisher can later Accept or Reject
- **Live** → credits awarded to publisher after verification

7.3 Link Verification

Triggered when a request is marked Live:

1. Platform headless crawler fetches `source_url`
2. Searches page HTML for a link with text matching `anchor_text` pointing to `target_url`
3. Records result in `link_verifications` audit log
4. **On success:** status = Live, credits awarded to publisher
5. **On failure:** user notified, no credits awarded, retry allowed (max 3 times)
6. After 3 failures: flagged for manual review

7.4 Domain Ownership Verification

Meta Tag method:

```
<meta name="blp-verify" content="{unique_token}">
```

Platform fetches homepage, parses `<head>`, confirms token match.

DNS TXT method:

```
TXT blp-verify.{domain} = {unique_token}
```

Platform performs DNS TXT lookup and confirms token.

Tokens are UUIDs generated per project. Verification is permanent unless domain ownership changes.

7.5 Responsiveness Score Formula

```
account_score = (incoming requests where status ≠ 'Pending') / total_incoming × 100
project_score = same formula scoped to a single project domain
```

Recomputed on every incoming request status change. Cached for display, refreshed within 24 hours. Auto-expired requests (no action after 30 days) count as non-responded and reduce the score.

8. Non-Functional Requirements

8.1 Performance

- Exchange browser loads in < 1.5s (server-side pagination, cached metrics)
- Request list loads in < 1s
- Link verification job completes within 60s
- Credit deduction is atomic (no race condition on concurrent requests)

8.2 Scalability

- Stateless API servers horizontally scalable behind a load balancer
- Background jobs via BullMQ + Redis
- Database connection pooling (PgBouncer)
- Read replicas for exchange browser queries

8.3 Security

- Passwords hashed with bcrypt (cost factor 12)
- JWT signed with RS256, 15-minute access token TTL
- Refresh tokens httpOnly + Secure + SameSite=Strict, 30-day TTL
- CSRF protection on all state-changing endpoints
- SQL injection prevented via parameterised queries
- Rate limiting on all auth and action endpoints (Redis-backed)
- Stripe webhook signature verification
- Domain verification tokens are UUIDs — unguessable

8.4 Reliability

- 99.9% uptime SLA
- Background job retries with exponential backoff (max 3 retries)
- Stripe payment events idempotently processed
- Credit transactions use database-level transactions (no double-spend)

8.5 Accessibility

- WCAG 2.1 AA compliance
- Keyboard navigable throughout
- ARIA labels on all icon-only buttons and tooltips
- Sufficient colour contrast for all status badges and score colours

8.6 Internationalisation

- UI in English (v1)
- Data layer supports multi-language content (language field on requests/websites)
- Dates in ISO 8601; displayed in user's local timezone

9. Information Architecture

/	My Projects (home)
/dashboard	Project Dashboard
/add-project	Add New Project
/exchange	
/websites	Browse Exchange Sites
/incoming	Incoming Requests
/outgoing	Outgoing Requests
/suggestions	AI-Powered Suggestions (Phase 3)
/marketplace	
/websites	Browse Marketplace Listings
/orders	My Orders
/transactions	Credit Transaction History
/credits/add	Purchase Credits
/profile	User Profile & Stats
/settings	Account Settings
/upgrade	Plans & Pricing
/help	Help Centre

10. Data Model

Table	Purpose
users	Account, profile, credit balance, subscription plan
projects	User's websites with verification and exchange status
project_metrics	Historical SEO metrics per project (DA, DR, TF, Traffic, RD, Spam)
websites	Exchange and marketplace inventory
website_metrics	SEO metrics for exchange/marketplace sites
backlink_requests	All exchange requests with full lifecycle fields
credit_transactions	Full double-entry ledger of credit movements
payments	Stripe payment records (credit purchases + marketplace orders)
marketplace_orders	Paid backlink order details and fulfilment status
responsiveness_scores	Cached per-user and per-project responsiveness scores
link_verifications	Audit log of automated live-link crawl checks
notifications	In-app notification queue

11. Backend Architecture

Tech Stack

Layer	Technology
Runtime	Node.js + TypeScript
Framework	Fastify
Database	PostgreSQL
Cache + Queue	Redis + BullMQ
Auth	JWT (RS256) + httpOnly refresh tokens
Payments	Stripe
Email	Resend
File Storage	AWS S3 (avatars, exports)
SEO Metrics	Moz API + Ahrefs API
Link Crawler	Puppeteer (headless Chrome)

API Surface (summary)

Domain	Key Endpoints
Auth	register, login, logout, refresh, forgot/reset password
Users	profile GET/PATCH, avatar upload, account stats, delete
Projects	CRUD, verify ownership, metrics + deltas, exchange toggle
Requests	list (incoming/outgoing), send, update status, mark live, export
Exchange	browse websites, website detail, export
Marketplace	browse listings, purchase, order history
Credits	balance, transactions, purchase (Stripe)
Responsiveness	account score, per-project score, public per-domain score
Webhooks	Stripe events
Admin	user management, listing management, credit adjustments

Background Jobs

Job	Trigger
Refresh project SEO metrics	Nightly cron
Refresh exchange/marketplace metrics	Nightly cron
Verify domain ownership	On-demand (user initiates)

Verify link is live (crawler)	On-demand (Mark as Live action)
Recompute responsiveness scores	On every status change
Auto-expire stale Pending requests	Daily cron (30-day threshold)
Send email notifications	Event-driven

12. Monetization

12.1 Credits (Primary Revenue)

Package	Credits	Price	Per Credit
Starter	5	\$9.99	\$2.00
Growth	15	\$24.99	\$1.67
Pro	35	\$49.99	\$1.43
Agency	80	\$99.99	\$1.25

Credits are consumed to send requests. Publishers earn credits back by accepting and publishing links. Net consumption drives repeat purchases.

12.2 Marketplace (Secondary)

Publishers list their site at a fixed USD price. Buyers pay directly. Platform takes a 15–20% commission per transaction.

12.3 Subscriptions (Phase 2)

Plan	Price	Key Limits
Free	\$0/month	3 projects, 10 requests/month
Pro	\$29/month	15 projects, unlimited requests, advanced filters
Agency	\$99/month	Unlimited projects, team members, white-label CSV exports

13. Launch Phases

Phase 1 — MVP (Months 1–2)

- Email/password auth
- Projects: add, verify (meta tag), exchange toggle, guidelines
- Exchange browser (seeded website list)
- Send / Accept / Reject / Hold / Mark Live (manual, no crawler)
- Credits: Stripe purchase, deduct on send, earn on live, transaction history
- Email notifications: new request, accepted, rejected, live
- Responsiveness score (computed live)

Phase 2 — Automation (Months 3–4)

- DNS verification method
- Automated link verification via headless crawler
- SEO metric auto-refresh (Moz/Ahrefs API integration)
- Responsiveness score persistence + auto-expiry (30-day Pending threshold)
- In-app notifications (real-time via WebSocket/SSE)
- Request and website list export (CSV, TSV, JSON, XLS)
- Dashboard metric deltas with date range picker

Phase 3 — Growth (Months 5–6)

- Marketplace (paid listings, Stripe Checkout, order tracking)
- Subscription plans (Pro, Agency via Stripe Billing)
- Google OAuth
- AI-powered site suggestions (match projects to exchange candidates by niche)
- Public profile pages with responsiveness score
- Admin dashboard
- Weekly email digest

14. Open Questions

#	Question	Owner
OQ1	Which SEO API provider — Moz, Ahrefs, or a unified aggregator? Cost vs. freshness trade-off.	Engineering
OQ2	How to handle link verification on JS-rendered (SPA) publisher sites? Puppeteer adds latency and cost.	Engineering
OQ3	Should credits be transferable between users (gifting/resale)?	Product
OQ4	What happens if a verified live link is later removed? Clawback policy needed.	Product
OQ5	Marketplace platform commission — 15% or 20%?	Business
OQ6	Should the exchange be open (any site joins) or approval-gated to maintain quality?	Product
OQ7	Free plan limits — are 3 projects and 10 requests/month the right thresholds?	Product
OQ8	Is responsiveness score fully public or only visible to matched exchange partners?	Product
OQ9	Mobile app — needed in Phase 1 or web-only sufficient?	Product
OQ10	Team/multi-user accounts — required before Agency plan launch?	Engineering